The examination is being carried out on the **following application documents**

Description, Pages

1-94 as originally filed

Sequence listings, SEQ ID NO

1-1131 as originally filed

Claims, Numbers

1-12 as originally filed

Drawings, Sheets

1/6-6/6 as originally filed

1 **Reference is made to the following documents ; the numbering will be adhered to in the rest of the procedure :**

D1: Linhart B. et al. : "Molecular approaches for diagnosis, therapy and prevention of cow's milk allergy", Nutrients, vol. 11, no. 7, 1492, 29 June 2019, pages 1-24

D2: Hochwallner H. et al. : "Cow's milk allergy: From allergens to new forms of diagnosis, therapy and prevention", Methods, vol. 66, no. 1, March 2014, pages 22-33

D3: WO 2015/048339 A2, 2 April 2015

2 **Unity - Art. 82 EPC :**

2.1 The present application discloses the identification of antigens responsible of allergy and their use for diagnosis purposes.

In the light of the prior art documents, the problem to be solved by the present application can be seen in providing alternative compositions and methods for diagnosing an allergy.
The polypeptides defined by (E1) to (E25) in claim 1 provide 13 solutions to the problem posed.

The common matter which can be identified as linking together the various polypeptides is that as they are identified as being milk derived antigens.

However, milk-derived antigens are known from the prior art. D1 discloses the identification of milk allergens, including among others caseins, α-lactalbumin, β-lactoglobulin and lactoferrin and recombinant epitopes thereof (Figure 1, Table 1) as well as hypoallergenic milk formulas. The alternative allergens of claim 1 only have in common that they are milk allergens. As this common feature has already been disclosed in the prior art it cannot be regarded as the special technical feature providing unity among said alternatives.
Hence, the above identified common general concept is not new and thus cannot be the single general inventive concept as required by Art. 82 EPC. The present application is therefore considered not to fulfil the requirements of unity as laid down in Art. 82 EPC. Therefore the claimed antigens according to the identified 13 groups of invention represent a distinct solution to the technical problem of providing further compositions and methods for diagnosing an allergy.

Moreover, the mere fact that the proteins are derived from the same source, namely milk is not sufficient to meet the criteria for unity of invention.
Due to the essential differences in the antigens contemplated as markers, which do not share a common structure, property or activity, no other technical features could be identified that form a technical relationship among each of the separate inventions claimed and which could be considered as corresponding special technical features in the sense of Rule 44(1) EPC.

2.2 Thus (E1)-(E25) define 13 inventions grouped according to the grouping disclosed in Table 2 and on p. 25 of the description which indicate that spots 2 and 3 are derived from the same protein.

Hence, it is considered that the present application contains 13 inventions which are not linked by a single general inventive concept in the sense of Art. 82 EPC as follows :

Invention 1 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E1) and (E2) belonging to immunoglobulin M heavy chain secretory form.

Invention 2 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E3), (E4), (E5), (E6), (E7) and (E8) belonging to β-1 metal binding globulin.

Invention 3 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E9) and (E10) belonging to lactoperoxidase.

Invention 4 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E11) and (E12) belonging to polymeric immunoglobulin receptor.

Invention 5 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E13) belonging to C3-beta-c.

Invention 6 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E14) and (E15) belonging to Ig heavy chain precursor (B/MT.4A. 17. H5. A5).

Invention 7 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E16) and (E17) belonging to perilipin.

Invention 8 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E18) and (E19) belonging to zinc-alpha-2-glycoprotein.

Invention 9 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E20) and (E21) belonging to glycoprotein 2.

Invention 10 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E22) belonging to lactadherin.

Invention 11 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E23) belonging to immunoglobulin light chain, lambda gene cluster.

Invention 12 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E24) belonging to intracellular cholesterol transporter 2.

Invention 13 : claims 1-12 (all in part)
compositions and methods for diagnosing an allergy comprising a polypeptide as defined in (E25) belonging to apolipoprotein A-IV.

2.3 The search has been limited to the invention first mentioned in the claims, namely invention 1 as defined herein above.

3 Clarity, support and disclosure - Art. 84 and Art. 83 EPC :

3.1 The set of claims relates to an allergy antigens and polypeptides whereas the description refers to *milk* allergy antigens. This inconsistency between the claims and the description contravenes Art. 84 EPC.

3.2 Claim 1 refers to a "kit for ... comprising ...." and claim 2 refers to a "composition for ... comprising ...". This must be construed as meaning merely a kit / composition suitable for ... . Indeed, the intended use of a product is not a technical feature of the product *per se* (Guidelines F-IV, 4.13).
A product which is *prima facie* identical to the products defined in a claim deprives said claim of novelty, even if said product has never been described for that use (Guidelines F-IV, 4.13), *i.e.*, the identification of a property which is inherent to a product is not sufficient to confer novelty to the product as such.

The same comment holds true for the antigen as defined in claim 4, the composition comprising the same according to claim 5, the composition of claim 6 and the "tester composition" of claim 9, which therefore all relate to the same subject-matter, namely only as comprising a polypeptide as defined in claim 1.

Thus claims 1, 2, 4-6 and 9 relate to the same subject-matter, which is further contrary to the requirements of clarity/conciseness under Art. 84 EPC as the use of several independent claims covering the same subject-matter renders the set of claims taken as a whole unclear and introduces a lack of conciseness (Art. 84 EPC ; Guidelines F-IV, 5).

3.3 The intended medical use of claim 6 is not limiting because claim 6 does not follow the wording recommended for claims according to Art. 54(5) EPC (Guidelines G-VI, 7.1.2).

3.4 SEQ ID NOs: 27, 28, 32, 35, 37, 40, 42, 47, 48 (QNVYTCQVEH NKETF) are identical as well as SEQ ID NOs: 29-31 (SQNVYTC), SEQ ID NOs: 33, 43 (LSQNVYTC), SEQ ID NOs: 34, 36, 39 (HNKE), SEQ ID NOs: 45, 51, 57 (YTCQVEHNKE), SEQ ID NOs: 46, 53, 56, 58 (HNKETF), SEQ ID NOs: 59, 60, 68, 74, 76, 81 (TKSKYVTSSPAPEPQ), SEQ ID NOs: 63, 77, 94 (KSKYVTSSP), SEQ ID NOs: 65, 78, 83 (KSKYVTSS), SEQ ID NOs: 67, 88, 91 (YVTS), SEQ ID NOs: 70, 79, 84, 89, 90 (SKYVTS), SEQ ID NOs: 71, 86, 96 (KYVTS). All these redundancies and definitions of the same matter by different denominations render the set of claims unclear (Art. 84 EPC).

3.5 Claim 1 relates to sequences having a large proportion of residues which can be any ("X") amino acid (SEQ ID NOs : 38, 49, 50, 52, 54, 55, 61, 62, 64, 66, 69, 73, 82, 85, 92, 93, 97). Some of the claimed peptides have up to 80% of the amino acid residues that are undefined. This renders the scope of the claim unclear in the sense of Art. 84 EPC and represents an undue burden to the skilled person and lack of sufficiency of disclosure within the meaning of Art. 83 EPC. Any reference to peptides comprising "X" residues should be removed from the claims.

3.6 Claims 11-12 lack support and sufficient disclosure in the description contrary to the requirements of Art. 84 and Art. 83 EPC as no experimental data whatsoever are presented which provide guidance about how the raw material according to independent claim 11 is to be obtained nor about how the method of producing the same according to independent claim 12 is to be performed.

Moreover, failing to provide an example of the product and method claimed is contrary to the requirements of Rule 42(1)(e) EPC which states that the description shall describe in detail at least one way of carrying out the invention claimed (Guidelines F-III, 1).

furthermore defined by what it does not contain which is completely unclear.

As there appears to be no possibility of overcoming this objection by amendment, claims 11-12 should be deleted.

4 Novelty - Art. 52(1) and 54 EPC :

4.1 Claim 1 relates to polypeptides identified by SEQ ID NOs wherein only 2 amino acids are defined. Thus any polypeptide mentioned in any prior art falls within the scope of claim 1. Therefore for this reason alone, claims 1-2, 4-6 and 9 lack novelty.

4.2 Document D1 discloses molecular approaches for diagnosis, therapy and prevention of cow's milk allergy but does not mention IgMHC (relevant passages cited in the European search report).

Similarly, document D2 is a review about allergens, diagnosis, therapy and prevention of cow's milk allergy. Table 1 lists known cow's milk allergens including α-lactalbumin, β-Lactoglobulin, bovine serum albumin, *Bos immunoglobulins*, lactoferrin, αS1-casein, αS2-casein, β-casein and κ-casein, but does not mention IgMHC in particular (relevant passages cited in the European search report).

4.3 Claim 1 is directed to a polypeptide comprising any of the SEQ ID Nos : 27-97 and 1115 ((E1) and (E2)) and therefore encompasses the full length immunoglobulin M heavy chain secretory form or fragments thereof.

Document D3 discloses discloses formulations comprising isolated nutritive polypeptides, such as milk proteins. The formulation may comprise a nutritive polypeptide which is non-allergenic or has reduced allergenicity as compared to a reference nutritive polypeptide (relevant passages cited in the European

search report). The polypeptide of SEQ ID NO: 1043 presents 96.94% identity to SEQ ID NO: 2 of the present application, but includes with 100% identity all the polypeptides identified by (E1) and (E2).

Thus the products as defined in claims 1, 2, 4-6 and 9, as well as the processed product as defined in claim 11 lack novelty over D3.

5 Inventive step - Art. 52(1) and 56 EPC :

5.1 In light of any one of D1 and D2 taken as the closest prior art document, the problem to be solved by present claims 3, 7, 8 and 10 can be seen in providing alternative methods for diagnosing allergy or detecting an allergen.

The problem is solved in said claims by using a polypeptide as defined in (E1) or (E2) (claim 3) or using an antibody directed thereto (claims 7, 10) or using a primer to detect nucleic acid encoding said polypeptide (claims 8, 10).

5.2 As the polypeptides defined by (E1) and (E2) are known form D3, it is considered that the skilled person who wishes to detect any polypeptides defined by (E1() and (E2) would regard it a normal design procedure to generate antibodies and primers directed thereto and thus to combine all the features set out in claims 3, 7, 8 and 10. Thus, claims 3, 7, 8 and 10 do not involve an inventive step.